

CO2 AT3

- 1) 1 Identify and fix the error in the given SQL query that uses incorrect
- 2) JOIN syntax and returns duplicate results.
- 3) 2 The following sub-query returns incorrect results due to a missing
- 4) correlation. Debug and rewrite it correctly.
- 5) 3 A given SQL view definition causes errors when queried. Identify
- 6) the issue and rewrite a correct and optimized view.
- 7) 4 Debug the given relational algebra expression that produces
- 8) incorrect output for a natural join operation.
- 9) 5 The SQL query below violates referential integrity. Identify the
- 10) violation and modify the statements to enforce proper constraints.
- 11) 6 Given an inefficient SQL query with nested loops, rewrite it using
- 12) JOIN to optimize performance.
- 13) 7 The GROUP BY query below produces wrong aggregation results.
- 14) Debug the query and explain the error.
- 15) 8 Identify and fix the logical error in a given SQL UPDATE statement
- 16) that unintentionally modifies all rows in the table.
- 17) 9 Optimize the given multi-table SQL query by restructuring sub-
- 18) queries into JOINS and using appropriate indexes.
- 19) 10 Debug a given relational calculus expression that returns an empty
- 20) result set due to incorrect quantifier usage.

```

mysql> USE dbms_lab;
Database changed
mysql>
mysql> CREATE TABLE Department(
->     DeptID INT PRIMARY KEY,
->     DeptName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.11 sec)

mysql>
mysql> CREATE TABLE Employee(
->     EmpID INT PRIMARY KEY,
->     EmpName VARCHAR(50),
->     Salary INT,
->     DeptID INT,
->     FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> INSERT INTO Department VALUES
-> (1, 'HR'),
-> (2, 'IT'),
-> (3, 'Sales');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Employee VALUES
-> (101, 'Arun', 30000, 1),
-> (102, 'Bala', 40000, 2),
-> (103, 'Charan', 35000, 2),
-> (104, 'Divya', 28000, 3),
-> (105, 'Esha', 45000, 2);
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

```

```

mysql> SELECT EmpName, DeptName
-> FROM Employee
-> JOIN Department
-> WHERE Employee.DeptID = Department.DeptID;

```

EmpName	DeptName
Arun	HR
Bala	IT
Charan	IT
Esha	IT
Divya	Sales

5 rows in set (0.01 sec)

```

mysql> SELECT EmpName, Salary
-> FROM Employee E
-> WHERE Salary >
-> (
-> SELECT AVG(Salary)
-> FROM Employee
-> );

```

EmpName	Salary
Bala	40000
Esha	45000

2 rows in set (0.01 sec)

```

mysql> CREATE VIEW EmpDept AS
-> SELECT EmpName, DeptName
-> FROM Employee;
ERROR 1054 (42S22): Unknown column 'DeptName' in 'field list'

```

```
mysql> SELECT *
-> FROM Employee
-> NATURAL JOIN Department;
+-----+-----+-----+-----+-----+
| DeptID | EmpID | EmpName | Salary | DeptName |
+-----+-----+-----+-----+-----+
| 1 | 101 | Arun | 30000 | HR |
| 2 | 102 | Bala | 40000 | IT |
| 2 | 103 | Charan | 35000 | IT |
| 2 | 105 | Esha | 45000 | IT |
| 3 | 104 | Divya | 28000 | Sales |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> INSERT INTO Employee
-> VALUES
-> (106, 'Kumar', 32000, 10);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('dbms_lab`.`employee`, CONSTRAINT `employee_ibfk_1` FOREIGN KEY ('DeptID') REFERENCES `department` ('DeptID'))
mysql> SELECT EmpName
-> FROM Employee
-> WHERE DeptID =
-> (
-> SELECT DeptID
-> FROM Department
-> WHERE DeptName='IT'
-> );
+-----+
| EmpName |
+-----+
| Bala |
| Charan |
| Esha |
+-----+
```

```
3 rows in set (0.06 sec)

mysql> SELECT DeptID, EmpName, AVG(Salary)
-> FROM Employee
-> GROUP BY DeptID;
ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'dbms_lab.Employee.EmpName' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
mysql> UPDATE Employee
-> SET Salary = Salary + 5000;
Query OK, 5 rows affected (0.06 sec)
Rows matched: 5 Changed: 5 Warnings: 0

mysql> SELECT EmpName
-> FROM Employee
-> WHERE DeptID IN
-> (
-> SELECT DeptID
-> FROM Department
-> WHERE DeptName='IT'
-> );
+-----+
| EmpName |
+-----+
| Bala |
| Charan |
| Esha |
+-----+
3 rows in set (0.05 sec)

mysql> SELECT *
-> FROM Employee
-> WHERE Salary >
-> (
-> SELECT MAX(Salary)
-> FROM Employee
-> );
Empty set (0.00 sec)

mysql> |
```